Hydrothermal synthesis of MoS2 core-shell structures

"In hydrothermal process, can we engineer the shell thickness of MoS2 ?" *Generated by LeapSpace*

Yes — **MoS_2 shell thickness can be engineered in hydrothermal synthesis**, most directly by changing the **MoS_2 precursor concentration**, and in some systems also by adjusting **reaction time** and the **template/core conditions** that control growth. [114] [115] [116] [117]

What the evidence shows

Hydrothermal control lever	Effect on MoS_2 shell	Evidence
Precursor concentration	Shell thickness increases as more MoS_2 precursor is supplied; one hydrothermal $BiVO_4$@MoS_2 system tuned the shell to an optimum of **300 nm**. [114]	Strong direct evidence
MoS_2 amount / loading	In a hydrothermal MoS_2@Au core-shell system, the MoS_2 layer thickness was tunable from about **14.9 nm to 88.7 nm** by changing the amount of MoS_2 precursor. [115]	Strong direct evidence
Hydrothermal reaction time	Longer hydrothermal time can drive growth from solid/rough particles toward **core-shell** and finally **uniform hollow core-shell** MoS_2 microspheres. [116]	Direct structural control evidence
Template + hydrothermal conditions	Hydrothermal growth on structured templates can produce **radially stacked MoS_2 nanosheets** and well-developed shells, showing that shell architecture is controllable through the growth environment. [117] [118]	Supportive evidence

Practical takeaway

If your goal is **thickness control**, the most defensible hydrothermal knobs are:

1. **Increase or decrease MoS_2 precursor concentration** to tune shell build-up. [114] [115]
2. **Adjust hydrothermal time** to move from thin/incipient coating toward thicker, more complete shells. [116]
3. **Use a template/core** that confines nucleation and limits uncontrolled lateral growth. [117] [118]

Bottom line

Yes — hydrothermal synthesis can engineer MoS_2 shell thickness, and precursor concentration is the clearest control parameter in the retrieved evidence. Reaction time and the choice of template/core also matter, but the exact thickness window depends on the specific core-shell system. [114] [115] [116]

Confidence level: **High —** multiple retrieved hydrothermal studies directly report tunable MoS_2 shell thickness or hydrothermally controlled shell formation.

References

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